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**COVID-19 Impacts on Anthropogenic NO_x Emissions in China, the contiguous
U.S., and Europe: Insights from Joint TROPOMI and Surface NO₂ Data
Assimilation**

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Abstract: This study integrates satellite and surface NO₂ measurements through the ensemble square root filter (EnSRF) and the GEOS-Chem model to estimate anthropogenic NO_x emissions for China, the contiguous U.S., and Europe over 2019–2020, and further examines how the COVID-19 pandemic altered emission patterns and sectoral contributions. Our results show that in 2020, anthropogenic NO_x emissions were 19.20 ± 5.87 Tg in China, 8.96 ± 2.90 Tg in the contiguous U.S., and 10.56 ± 2.95 Tg in Europe. Compared with 2019, emissions decreased by about 5.38% in China, 8.94% in the contiguous U.S., and 15.38% in Europe. In China, emission reductions came from transportation, power plants, and industry. In contrast, reductions in the contiguous U.S. and Europe were almost entirely from transportation. These differences reflect the fundamental gap in energy structure and industry models between developed and developing economies. During the first wave of the pandemic, emissions fell sharply compared to the same period in 2019. The decreases were 18.04% in China, 10.56% in the contiguous U.S., and 21.74% in Europe. Large spatial variations were observed within each region. This work provides observation-based evidence of NO_x emission changes across major world regions during COVID 19. It highlights how emissions can behave differently under sudden public health crises. The findings are useful for checking bottom-up emission inventories, assessing extreme events' impacts on air quality, and designing better emission reduction policies.

Keywords: NO_x inversion; GEOS-Chem; data assimilation; EnSRF; COVID-19

1. Introduction

Nitrogen oxides (NO_x = NO + NO₂) are key reactive components in tropospheric chemistry, playing important roles in ozone formation, hydroxyl radical cycling, and secondary aerosol production. Anthropogenic emissions from transportation, industry, and power dominate the global NO_x budget, affecting not only regional air quality and public health but also the climate system through altered atmospheric oxidizing capacity and radiative forcing [1-3]. Therefore, accurate quantification of anthropogenic NO_x emissions is essential for assessing emission reduction policies, validating bottom-up inventories, and understanding atmospheric responses to abrupt disturbances such as the COVID-19 pandemic.

Bottom-up emission inventories rely on activity data, emission factors, and spatial proxies. Despite improving spatial resolution, their temporal resolution generally remains monthly, making it difficult to capture emission variability at finer scales [4-6]. Furthermore, the gridding of global emissions,

incomplete regional statistical records, and the inappropriate use of emission factors further constrain inventory accuracy [7-9]. Meanwhile, the time lag in obtaining activity data inevitably reduces the timeliness of bottom-up emission inventories, hindering timely access to recent actual emissions and thus limiting their utility for near-real-time policy decisions and emergency responses [10-12]. This limitation is more evident during abrupt events such as the COVID-19 pandemic [13-16].

In contrast, top-down methods invert NO_x emissions from observed NO_2 concentrations, reducing inventory uncertainties and enabling dynamic emission estimates [17-21]. Mass-balance-based inversion methods assume a linear relationship between local NO_x emissions and observed NO_2 concentrations, thereby simplifying the treatment of atmospheric transport and nonlinear chemical processes, which allows NO_x emissions to be retrieved at different spatial scales with relatively low computational cost [20, 22-24]. However, this simplification of physical and chemical processes also introduces substantial uncertainties in emission retrievals under complex conditions, such as emission smearing errors [22, 25-27]. In comparison, data-assimilation-based inversion methods account more comprehensively for the effects of atmospheric transport and nonlinear chemical processes. By minimizing the discrepancy between observations and model simulations, they achieve higher-precision retrievals of anthropogenic NO_x emissions at regional to global scales [13, 16, 18, 19, 28, 29].

The COVID-19 pandemic led to unprecedented changes in human activities and emissions in early 2020, with a sharp decline over a short period [30-32]. China, the U.S., and Europe dominate global anthropogenic emissions yet differ fundamentally in industrial structure, energy consumption, and environmental policy, making a comparative investigation of their COVID-19 emission responses essential for understanding energy adjustments, emission resilience, and policy effectiveness under extreme perturbations. However, studies based on data assimilation methods to examine the impact of COVID-19 on emissions remain relatively scarce, and most are limited to analyzing emission changes in a single region, lacking a systematic comparison of China, the U.S., and Europe within the same inversion framework, as well as a discussion of sectoral differences in emission changes across these regions [13, 16, 33, 34]. Moreover, most NO_x emission inversion studies use only a single observational source, either satellite [16, 18, 28] or surface measurements [13, 35], while studies combining both remain limited. The wide-area coverage of satellite observations and the high precision of surface measurements complement each other, and their integration is expected to provide stronger constraints on NO_x emission inversions [36-39]. The added value of joint assimilation over single-source inversions has been quantified in our previous work [39], revealing that inversion results derived from a single observational source are susceptible to seasonal and regional factors, and confirm that the fusion of

satellite and surface data effectively improves the robustness of emission estimates. However, a systematic evaluation of this framework under long-term continuous operation and extreme disturbance conditions remains absent.

To address the aforementioned research gaps, this study employs a top-down emission inversion framework based on the ensemble square root filter (EnSRF) data assimilation method and the GEOS-Chem nested model [39]. Building upon the method validation presented in our previous study [39], the present work shifts the research focus from the relative merits of single- versus multi-source assimilation to the robustness and applicability of the joint framework over extended temporal scales and under abrupt emission perturbations. For the three major regions of China, the United States, and Europe, we jointly assimilate TROPOMI NO₂ tropospheric vertical column densities and surface NO₂ observations from each region, aiming to estimate anthropogenic NO_x emissions independently for each region during 2019 and 2020. This two-year continuous inversion spans both a normal emission year (2019) and an anomalous period marked by the COVID-19 lockdowns (2020), thereby enabling an evaluation of the framework's capability to track stable baseline emissions as well as rapid, sector-specific perturbations. Using 2019 as the baseline year, we investigate the impacts of the COVID-19 pandemic on emission patterns and emission sectors across different regions.

2. Method and Data

2.1 Daily emission inversion framework

The anthropogenic NO_x emission inversion framework used in this study is consistent with that described in Wu et al [39]. This framework is based on the nested version of GEOS-Chem 14.3.0 [40], employing a full-chemistry simulation scheme with a horizontal resolution of $0.5^\circ \times 0.625^\circ$ and 47 vertical layers, driven by the MERRA-2 reanalysis dataset [41]. The prior anthropogenic emissions are obtained from the EDGAR v8.1 inventory [42]. Following previous studies, the prior emission uncertainty was prescribed as 40%. This value is widely adopted in NO_x assimilation frameworks and provides a reasonable range of prior errors [21, 29, 43]. Based on observing system simulation experiments (OSSEs; see Text S1), the ensemble size was set to 40 and the localization lengths to 210 km for China and Europe and 180 km for the contiguous U.S. The OSSEs indicated that a 40-member ensemble provides a good balance between inversion performance and computational cost and that the selected localization lengths are optimal. Further details of the inversion framework can be found in Text S2 and our previous study [39].

In this study, GEOS-Chem was spun-up from July 1, 2018, to December 29, 2018, to obtain the

initial model fields. The emission inversion framework was run from December 29, 2018, to January 1, 2021. The first three days of the inversion served as the spin-up phase and were excluded from the results.

2.2 Satellite and surface observation data

The TROPOspheric Monitoring Instrument (TROPOMI) is a shortwave spectrometer jointly developed by the Netherlands and the European Space Agency. This study employs the TROPOMI tropospheric NO₂ Level2 reprocessed product version 2.4.0. To ensure the quality of assimilated data, we only consider observations not affected by clouds and snow/ice cover ($qa_value > 0.75$). The simulated satellite NO₂ vertical column density is derived by applying the TROPOMI averaging kernel as $N_v^{trop,model} = \sum_l A_l^{trop} sim_l$, where A_l^{trop} is the TROPOMI tropospheric averaging kernel in layer l , sim_l is the GEOS-Chem partial column in layer l [44].

Hourly surface NO₂ concentration observations are obtained from the China National Environmental Monitoring Centre (CNEMC), the European Environment Agency (EEA), and the U.S. Environmental Protection Agency (EPA) Air Quality System (AQS). For surface observations, we applied a series of quality control procedures and retained approximately 10% of the stations for independent validation. In addition, surface NO₂ concentrations were corrected following the recommendation of Lamsal et al [45]. Further details can be found in Supplementary Text S3.

At the $0.5^\circ \times 0.625^\circ$ resolution, individual grid cells typically encompass multiple ground-based stations and satellite pixels with highly correlated observational biases. Assimilating these observations individually tends to produce redundant adjustments in dense regions [13], introduce unresolved small-scale noise [43, 46, 47], and substantially increase computational costs [43, 48]. To address these issues, this study constructs super-observations for both satellite and surface observations to reduce scale mismatches, improve observational representativeness, and optimize assimilation efficiency. The methods used to construct satellite and surface super-observations, along with their observation errors, are described in our previous study [39].

3. Results and Discussion

3.1 Changes in total emissions between 2019 and 2020

Prior to quantifying emissions, we evaluated the simulated NO₂ concentrations against satellite (Figures S1–S2, Table S1) and surface (Figures S3–S6, Tables S2–S3) observations, thereby providing indirect support for the reliability of the inversion results. Details are provided in Text S4.

Based on the inversion results and the sectoral emission fractions from the EDGAR inventory,

Figure 1 presents the year-on-year changes in emissions and their spatial distribution differences for China, the contiguous U.S., and Europe in 2020 relative to 2019. In 2020, China's anthropogenic NO_x emissions were estimated at 19.20 ± 5.87 Tg, representing a decrease of approximately 5.38% from 2019 (20.29 ± 6.06 Tg) (Figure 1a). Sectoral attribution based on prior EDGAR fractions suggests that the transportation, power generation, and fuel exploitation sectors were the principal contributors to this reduction, accounting for 44.95%, 30.28%, and 19.27% of the total reduction, respectively. The transportation sector's decline occurred during the initial lockdown period (January–March 2020), a period coinciding with substantial reductions in mobility and economic activity. The power sector's reduction may be associated with two factors: first, the continued advancement of ultra-low emission retrofits in coal-fired power plants, coupled with a declining share of coal-fired electricity and an increasing proportion of non-fossil energy sources; second, the reduction in industrial electricity demand coincident with COVID-19 [23, 49, 50]. In terms of spatial changes (Figure 1c), emissions across Chinese regions in 2020 were generally lower than in 2019, with the North China Plain (regional divisions shown in Figure S7) exhibiting the largest reduction magnitude (Figure S8). Regarding sectoral contributions, based on prior EDGAR fractions, the transportation and power sectors were the principal sectoral contributors to emission reductions, jointly accounting for over 60% of the total reduction (Figure S9). The southeastern region, as one of China's major economic zones, recorded a reduction of merely 4.79%, which may reflect a transient overshoot in manufacturing output following the lifting of lockdown measures, with the industrial combustion sector offsetting approximately 22% of the overall reduction based on prior sectoral fractions. Northeast China, distant from the pandemic epicenters, experienced a relatively modest decline in the transportation sector compared to other regions; meanwhile, as a significant industrial base, the region saw a 4.20% year-on-year increase in industrial combustion emissions. With these combined sectoral contributions, the overall reduction in Northeast China was only 2.84%, a smaller reduction than in other regions, consistent with findings from previous studies reporting relatively small reduction magnitudes in this region [24, 51].

In 2020, anthropogenic NO_x emissions in the contiguous U.S. and Europe were estimated at 8.96 ± 2.90 Tg and 10.56 ± 2.95 Tg, respectively, representing decreases of approximately 8.94% and 15.38% from 2019 (9.84 ± 3.23 Tg and 12.48 ± 3.38 Tg) (Figures 1d and 1g). These reduction magnitudes are broadly consistent with reports from the U.S. Environmental Protection Agency [52] and the European Environment Agency [53]. In contrast to China, whose economic structure and energy consumption patterns differ, the U.S. and Europe, as developed regions, have largely offshored their industrial manufacturing, with anthropogenic NO_x emissions being dominated by the transportation sector (Figure

S10). Correspondingly, sectoral attribution based on prior EDGAR fractions suggests that emission reductions in 2020 were also predominantly contributed by this sector, contributing 79.55% and 79.69% of total reductions in the contiguous U.S. and Europe, respectively. Spatially (Figures 1f and 1i), NO_x emissions across all the contiguous U.S. and European regions in 2020 declined significantly relative to 2019 coincident with COVID-19. In the contiguous U.S., the New England, Middle Atlantic, Mountain, and Pacific regions exhibited the largest reductions, all exceeding 10% (Figure S8), with sectoral attribution suggesting that the transportation sector was also the dominant contributor (Figure S11). In Europe, the western and southern regions showed the most pronounced declines of approximately 20%, with sectoral attribution suggesting that the transportation sector contributed over 80% of the reductions (Figure S12). Notably, the East South Central and West South Central regions of the contiguous U.S. experienced smaller reductions than other regions, with the West South Central region recording the lowest decline at merely 2.77% year-on-year. This disparity may be associated with two factors. First, industrial combustion emissions in these two regions increased in 2020; second, the reduction magnitude in their transportation sectors was relatively modest compared to other regions. These characteristics are consistent with these areas having been comparatively less affected by the COVID-19 pandemic.

A comparison of the inversion results from this study with the EDGAR v8.1 prior emission inventory, the CAMS v6.2 global emission inventory [54], the MEIC v2.0 Chinese emission inventory [55, 56], satellite-based mass-balance inversion estimates for China, the contiguous U.S., and Europe [24], and ground-based observation-assimilated inversion estimates for China [29] is presented in Figure S13. The results indicate that, within the range of inversion uncertainties (one standard deviation of ensemble members), the total emissions inverted in this study show reasonable consistency with other research findings. The prior emission inventory slightly overestimates China's emissions while underestimating those of the contiguous U.S. and Europe. It should be noted that the European emissions inverted in this study are higher than other estimates, which may be attributable to the assimilation of all-category surface observations across Europe. Specifically, point-source observations such as traffic stations, which carry strong local signals, are directly assimilated into the atmospheric chemical transport model at $0.5^{\circ} \times 0.625^{\circ}$ resolution, biasing the inversion results toward local emission hotspots [39]. However, this bias is expected to be systematic, the relative changes inferred from our analysis remain reliable and meaningful. Regarding relative emission changes, the inverted Chinese emission change in this study falls within the range of other estimates (from 1.06% to -10.22%), aligning closely with the MEIC inventory (-3.88%). The contiguous U.S. emission change also falls within the range of

other results (−3.88% to −16.12%), approximating the EDGAR inventory (−10.74%). The European emission change is slightly higher than other estimates (−5.96% to −13.54%), but remains reasonably close to the EDGAR inventory (−13.54%) and the CAMS inventory (−13.47%).

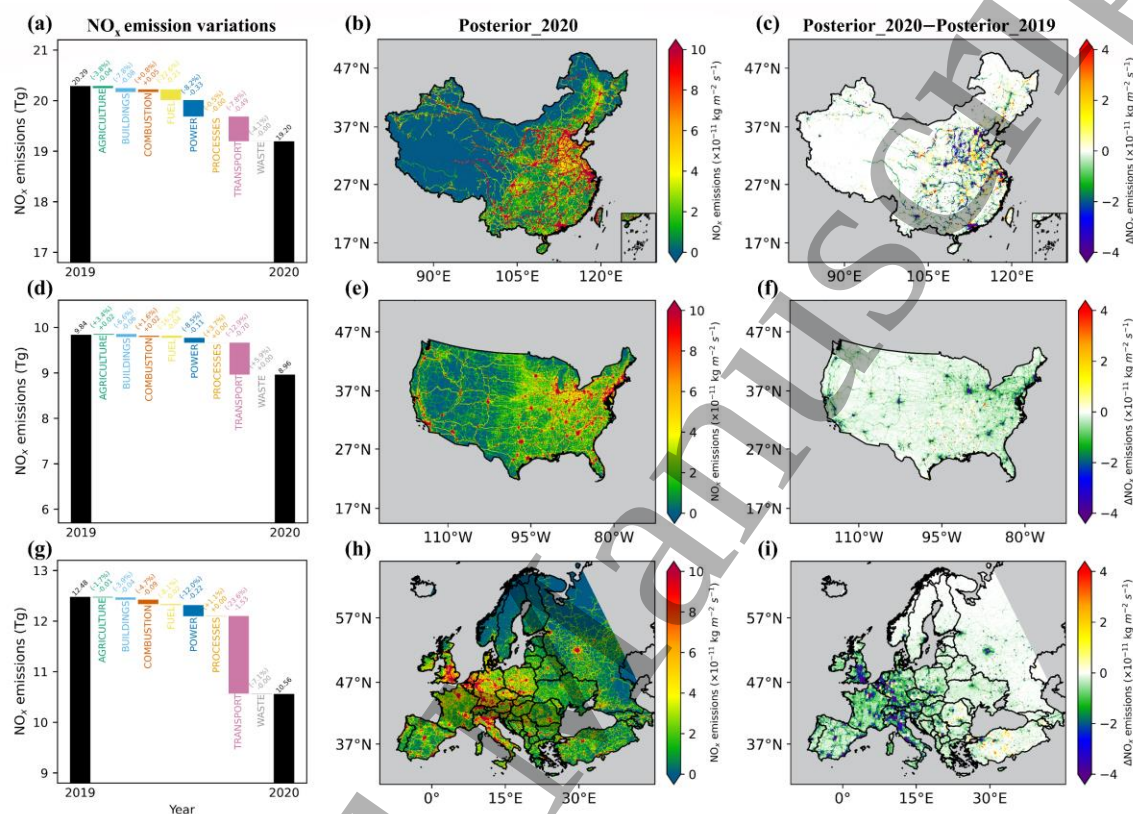


Figure 1. Anthropogenic NO_x emission changes in (a–c) China, (d–f) the contiguous U.S., and (g–i) Europe during 2019–2020. (a, d, g) Year-on-year changes in sectoral emissions, with values indicating absolute emission reductions and relative emission changes (in parentheses) for each sector; (b, e, h) spatial distributions of posterior anthropogenic emissions in 2020; (c, f, i) posterior emission differences in 2020 relative to 2019.

3.2 Monthly Variations in Emissions from 2019 to 2020

In terms of monthly emission variations (left panels of Figure 2), although the estimated emissions for China, the contiguous U.S., and Europe differ across methodologies, most approaches successfully reproduce the seasonal emission pattern characterized by higher emissions in cold seasons and lower emissions in warm seasons, consistent with seasonal heating demand patterns across all three regions. Notably, the CAMS emission inventory deviates from other estimates for the contiguous U.S. summer, showing enhanced summer emissions, which may reflect peak monthly allocation factors during summer [57]. Furthermore, most results adequately capture the emission trough in China during January–February, consistent with the Spring Festival effect. Regarding month-to-month emission

changes in 2020 relative to 2019 (right panels of Figure 2), most emission estimates successfully capture the substantial year-on-year emission declines in China during January–March that were coincident with the outbreak and lockdown measures. During April–June, only the inversion results from this study, the CAMS emission inventory, and Mao et al [24] capture an emission rebound consistent with the late lockdown and post-lockdown periods [20, 23]. However, in December, the emission changes estimated by Mao et al [24] and Kong et al [29] diverge considerably, which may arise from several factors. First, Mao et al [24] based their inversion on satellite columns, which are sensitive to the entire atmospheric NO₂ column; in winter, TROPOMI data are subject to increased observational uncertainties due to snow cover and cloudiness compared with summer conditions. Kong et al [29] performed inversions using surface observations, with constraints concentrated near the surface layer. This fundamental difference in observational representativeness may lead to biases in emission estimates. Second, the two studies differ in their inversion methodologies, prior emission inventories, and model resolutions, all of which can affect posterior emission results. In addition, differences in spatial resolution may also introduce varying degrees of estimation errors in emission hotspots [58]. In contrast, the present study, at a relatively finer spatial resolution, jointly assimilates TROPOMI columns and surface NO₂ measurements, thereby incorporating observational information from both the free troposphere and the surface layer, while the assimilation algorithm balances the gain weights of the two observation types based on their respective observation error covariances. This multi-source joint constraint mechanism can suppress biases arising from the insufficient representativeness of a single observational source. Our joint-assimilation estimate of the December year-over-year change falls within the range spanned by the two single-source inversions, offering a reconciled quantitative reference. In the contiguous U.S., both the inversion results from this study and Mao et al [24] satisfactorily reproduce the sharp emission declines in March and October coincident with COVID-19-related disruptions, whereas the CAMS emission inventory exhibits pronounced deviations from April onward. In Europe, all emission estimates consistently capture the significant reductions during March–May coincident with COVID-19; specifically, the reduction magnitudes estimated by this study and Mao et al [24] are relatively close in March, while the reduction magnitudes in April–May align more closely with the CAMS emission inventory. Following the resurgence of the European outbreak in October–November, the inversion results from this study and the CAMS inventory both clearly capture the emission declines during this period, consistent with renewed restrictions.

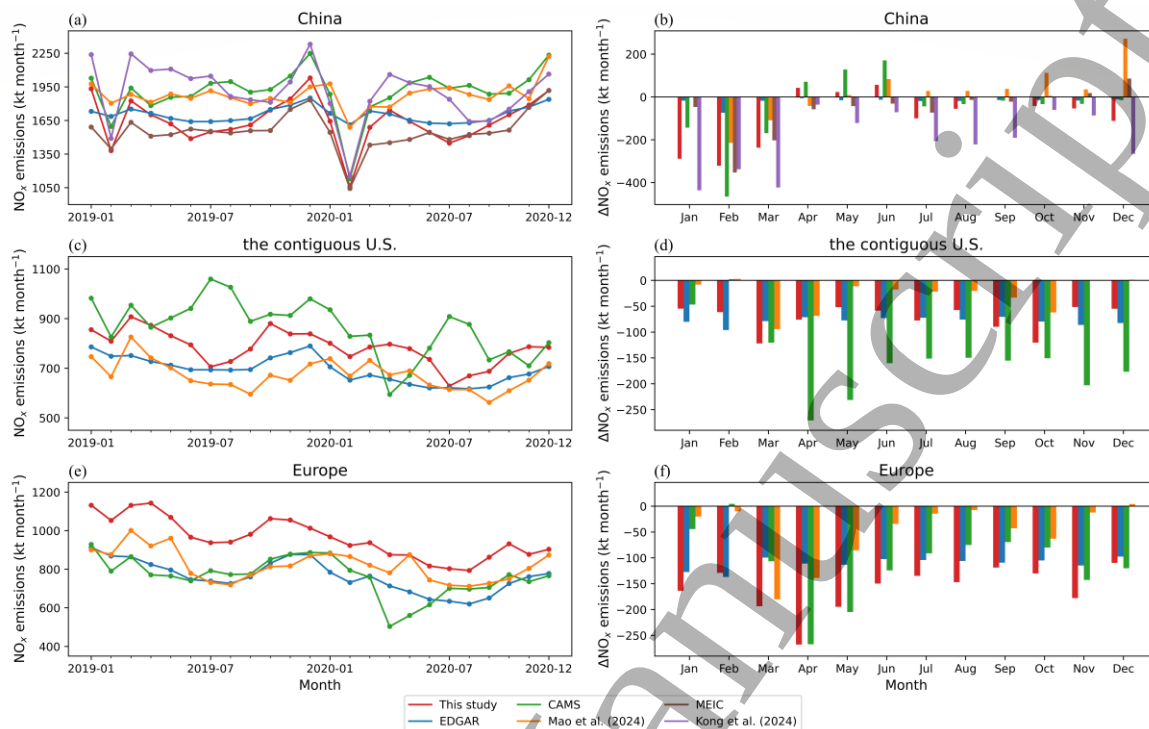


Figure 2. Comparison of posterior emissions with different emission estimates for (a–b) China, (c–d) the contiguous U.S., and (e–f) Europe. Left panels: monthly emission time series for 2019–2020; right panels: month-to-month emission changes in 2020 relative to 2019.

To elucidate the spatial heterogeneity of emission changes and their sectoral driving mechanisms, this study further conducts sub-regional scale analyses, which resolve monthly emission changes in 2020 relative to 2019 and sectoral contributions for sub-regions across China, the contiguous U.S., and Europe. These sectoral contributions are derived by applying posterior total emissions to prior sectoral fractions and should be interpreted accordingly. In China (Figure S14), the North China Plain experienced total emission reductions of 53.06–125.84 kt during January–March, with transportation and industrial combustion as the primary drivers; followed by an emission rebound dominated by the industrial combustion sector during April–June. The southeastern region exhibited a characteristic pattern of deep declines during January–March followed by robust rebounds from April onward, with the industrial combustion sector rebounding by 22.7–29.2 kt. The prior inventory, constrained by fixed activity data, severely underestimated both the spring reduction magnitude and rebound intensity. For the North China Plain, the prior industrial combustion maintained positive changes of 4.90–11.93 kt throughout the year, while the prior industrial combustion in the southeastern region remained at approximately 16–17 kt, in stark contrast to the posterior reversal patterns. Furthermore, the inversion results captured an abrupt emission plunge in the northwestern region during July–August coincident with localized COVID-19 outbreaks, a signal that the prior failed to reproduce.

In Europe (Figure S15), southern Europe saw total emission reductions of 85.26–101.41 kt during March–April, with transportation contributing approximately 70%; the prior underestimated transportation by more than a factor of two. Western Europe experienced sustained large declines during March–May (–45.12 to –62.35 kt), with the transportation sector remaining the largest contributor (–38.03 to –45.07 kt), while the prior reduction magnitude was substantially lower than the posterior. It should be noted that the emission reduction in northern Europe during January was even greater than that during the peak COVID-19 period (March–May), with the power generation and non-industrial sectors also showing much larger reductions than in other months. This emission anomaly is likely associated with the anomalously warm temperatures in January 2020 relative to January 2019 (Table S4). The warmer weather may have reduced residential and commercial heating demand, potentially leading to substantial declines in non-industrial combustion sector emissions, and may also have decreased total demand for the power generation sector, thereby contributing to reduced emissions. In the contiguous U.S. (Figure S16), the overall changes were relatively modest, yet the Middle Atlantic, East North Central, and East South Central regions exhibited abrupt emission plunges during March–April (approximately –10 to –28 kt), while the Mountain and Pacific regions showed deeper reductions during August–October (approximately –19 to –25 kt), with sectoral attribution suggesting that the transportation sector remaining the principal contributor to emission reductions.

In summary, multi-source observation-based inversions directly constrain total 2020 anthropogenic NO_x emission changes, which show non-stationary fluctuations; applying posterior totals to prior sectoral fractions suggests that transportation, industrial combustion, and power generation were the dominant sectoral contributors. Prior inventories based on fixed activity levels appear unable to accurately capture the true emission magnitudes during periods coincident with lockdowns and subsequent economic recovery under complex abrupt public health events such as COVID-19.

3.3 Diurnal Emission Variations in Response to COVID-19

To quantify the response of daily emissions to pandemic control measures, it is necessary to define the lockdown period for each region during the first wave of the pandemic. The initial wave of the pandemic was characterized by abruptness and exogeneity, with its impact on human activity patterns occurring suddenly. At that time, countries had not yet established systematic normalized epidemic prevention mechanisms, and emission fluctuations were likely primarily associated with lockdown intensity, with relatively minor influence from subsequent policy combinations and public behavioral adaptation. Therefore, this study focuses on associations during the initial wave. Based on national policy milestones and reopening timelines, the initial wave periods are defined as follows. For China,

the initial wave period is defined as January 23 to April 8, spanning from the announcement of the Wuhan lockdown to its lifting. For the contiguous U.S., the initial wave period is defined as March 13 to May 15. March 13 marks the declaration of a national emergency, after which states successively issued stay-at-home orders. Given the heterogeneity in state-level reopening times, May 15 is selected as a representative cutoff date when most states had begun gradual economic restart. For Europe, the initial wave period is defined as March 10 to May 11. March 10 marks Italy's announcement of a nationwide lockdown, followed by successive restrictive measures in Germany, France, Spain, and other countries; May 11 marks the beginning of gradual reopening for most European countries. Emissions in 2019 represent the pre-pandemic baseline, serving as the reference for assessing emission changes in 2020 that were coincident with or potentially associated with COVID-19 containment measures in each region.

To disentangle the relative contributions of meteorological conditions and emission changes to variations in surface NO_2 concentrations during the pandemic period, this study designed four sets of sensitivity simulation experiments based on GEOS-Chem (Text S5). The baseline scenarios employ 2020 emissions with 2020 meteorology and 2019 emissions with 2019 meteorology, respectively, representing the actual change conditions; the meteorology-varied scenario holds 2019 emissions fixed while applying 2020 meteorology; the emission-varied scenario fixes 2019 meteorology while applying 2020 emissions. Through inter-comparison of experimental results, the NO_2 concentration changes during the pandemic period can be decomposed into emission-driven and meteorology-driven contributions. Simulation results indicate that, relative to the same period in 2019, surface NO_2 concentrations across sub-regions in China (Figure S17), Europe (Figure S18), and the contiguous U.S. (Figure S19) all exhibited varying degrees of reduction during the initial wave in 2020, with emission changes as the dominant driver of NO_2 concentration declines. Given the above validation of emission dominance, the following analysis further focuses on the response characteristics of anthropogenic NO_x emission inversions during the initial wave period.

In China, anthropogenic NO_x emissions during the COVID-19 pandemic period (gray-shaded in Figure 3) declined by 18.04% relative to the same period in 2019, with a mean daily emission reduction of 9.99 kt (Figures 3a and 3d), accounting for 70.40% of the total annual reduction. By sector, based on prior sectoral fractions, transportation, industrial combustion, and power generation appear to have been the principal contributors to the reduction, accounting for 29.87%, 27.53%, and 21.33% of the total reduction, respectively (Figure 3g). Total emissions rebounded rapidly from March 2020 onward, temporarily exceeding the 2019 levels following the lifting of lockdown measures, then gradually

recovering to levels comparable to 2019 after a brief period of pent-up demand-driven production recovery, although industrial combustion emissions remained elevated. It should be noted that Chinese emissions had already exhibited a declining trend before the implementation of pandemic lockdowns, primarily consistent with the Spring Festival effect, during which substantial industrial production declined and associated emissions decreased, contributing to a phased decline in anthropogenic NO_x emissions. To quantitatively assess the superimposed impact of the Spring Festival effect during the COVID-19 period, we compared the emission changes during the two-week periods before and after the Spring Festival in 2019 and 2020 (blue and red dashed lines in Figure 3). Prior to the Spring Festival, anthropogenic NO_x emissions in China were 771.80 kt and 754.75 kt in 2019 and 2020, respectively, indicating comparable emission levels and suggesting that the pandemic had not yet had a significant impact at that stage. Following the Spring Festival, emissions dropped to 622.92 kt and 514.57 kt, respectively, with 2020 emissions 108.35 kt lower than 2019. This period normally marks the gradual post-holiday resumption of work, but the substantially lower emissions in 2020 relative to 2019 may indicate that lockdown measures likely suppressed the normal recovery of industrial and socioeconomic activities. At the sub-regional scale (Figure S20), the initial wave of the pandemic caused substantial reductions in the North China Plain and the southeastern region, with anthropogenic NO_x emissions declining by 24.94% and 18.42%, respectively, relative to the same period in 2019. Based on prior sectoral fractions, industrial combustion, power generation, and transportation were the common principal sources of reduction in both regions; specifically, industrial combustion dominated the reduction in the North China Plain, while transportation contributed most prominently in the southeastern region. By contrast, the northwestern region, distant from the pandemic epicenters, experienced relatively minor impacts.

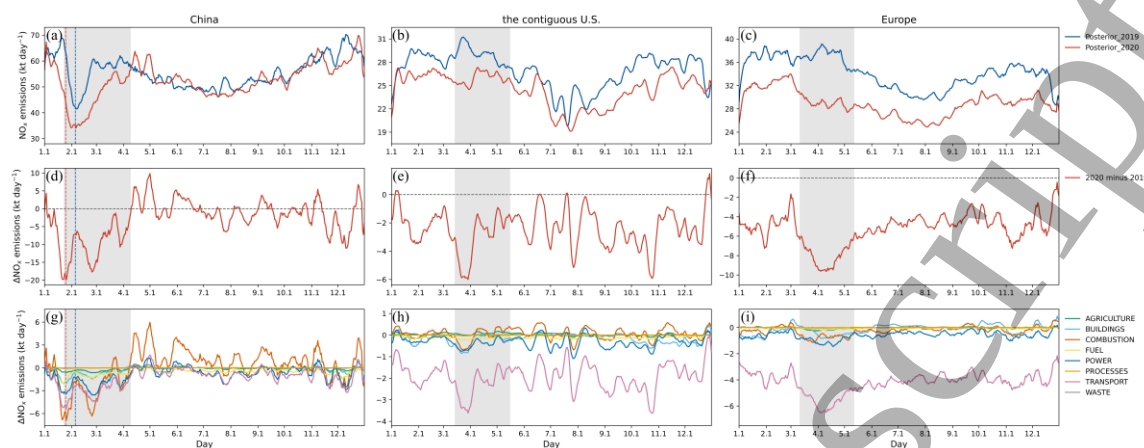


Figure 3. Seven-day moving averages of posterior emissions. (a–c) Daily comparisons between 2019 and 2020 for China, the contiguous U.S., and Europe; (d–f) daily anomalies in 2020 relative to 2019; (g–i) sectoral decomposition of the 2020–2019 differences. Blue and red dashed lines denote the Chinese Spring Festival in 2019 (February 5) and 2020 (January 25), respectively; gray shading indicates the initial COVID-19 lockdown period.

In the contiguous U.S., anthropogenic NO_x emissions during the COVID-19 pandemic period declined by 10.56% relative to the same period in 2019, with a mean daily emission reduction of 3.07 kt (Figures 3b and 3e), accounting for 22.38% of the total annual reduction. Based on prior sectoral fractions, transportation appears to have been the most prominent contributor to sectoral reductions, accounting for 74.87% of the total reduction (Figure 3h). It should be noted that although the U.S. states broadly resumed economic activity by May 15, emissions rebounded rapidly from April onward, recovering to pre-national-emergency levels by mid-April, with industrial combustion emissions even surpassing 2019 levels. At the sub-regional scale (Figure S21), the transportation sector served as the principal contributor to reductions across all regions, with most areas declining by approximately 13% relative to 2019. However, the Middle Atlantic and South Atlantic regions exhibited reduction magnitudes exceeding 19%, with total emissions declining by 18.70% and 14.58%, respectively, relative to 2019, consistent with particularly severe impacts on these two regions during the initial wave. The West North Central region experienced relatively minor changes, declining by only 5.30% relative to 2019.

In Europe, anthropogenic NO_x emissions during the COVID-19 pandemic period declined by 21.74% relative to the same period in 2019, with a mean daily emission reduction of 8.05 kt (Figures 3c and 3f), accounting for 26.44% of the total annual reduction. As in the contiguous U.S., based on prior sectoral fractions, the transportation sector in Europe appears to have been the principal contributor to reductions during this period, at 68.79% (Figure 3i). European emissions began rebounding from mid-April,

although at a more gradual pace than in China and the United States. Notably, despite the sequential implementation of pandemic response measures in Europe and the U.S. from mid-March, emissions had already exhibited discernible downward trends relative to 2019 from early March. This may suggest that human activities may have been affected by the pandemic prior to policy interventions, consistent with behavioral responses preceding policy mandates. At the sub-regional scale (Figure S22), the transportation sector served as the principal contributor to reductions across all regions; with the exception of Eastern Europe, all other sub-regions exhibited reduction magnitudes exceeding 23% relative to 2019. The initial wave coincided with the most severe emission changes in Southern Europe, with emissions declining by 35.09% relative to 2019, and reduction magnitudes across all emission sectors were markedly higher than in other sub-regions. Northern and Eastern Europe showed relatively weak pandemic responses, with emissions declining by 13.08% and 11.08%, respectively, relative to 2019.

It should be noted that the U.S. operates under a federal system with state-level autonomy, while European countries independently formulated their own epidemic prevention strategies; this decentralized decision-making model differs from China's unified prevention and control approach. Consequently, using relatively broad COVID-19 period definitions to assess sub-regional emission changes may obscure uncertainties arising from policy heterogeneity. Furthermore, the relatively lenient pandemic containment policies in Europe and the United States enabled the inversion results to capture a recurrence of emission trajectories similar to the initial wave from mid-October onward, with emissions declining significantly relative to 2019, and the rebound recovery speed in the U.S. continued to outpace that in Europe.

3.4 Uncertainties and Limitations

Although this study quantifies inversion uncertainties using ensemble methods, these uncertainties primarily reflect prior emission variability and model stochasticity. It should be emphasized that satellite retrieval errors, surface observation representativeness and measurement errors, and approximations in atmospheric chemical transport models regarding transport processes, vertical mixing, free-tropospheric NO₂ concentrations, and NO₂ chemical lifetimes all introduce additional errors into the inversion results. Furthermore, while the current horizontal resolution of 0.5° × 0.625° adequately represents chemical processes in strong emission source regions [58], further improvements in chemical feedback and an enhanced capability to capture local emission hotspots may be achieved by increasing the resolution to 5–15 km in the future [25, 59], thereby better constraining local emission signals.

The uncertainty in sectoral emission estimates stems from discrepancies in activity data and

emission factors across different inventories, resulting in distinct sectoral emission fractions characteristic of each inventory. In this study, sectoral emissions are derived from the ratios of grid-level sectoral emissions to total emissions in the EDGAR inventory; this reliance on a single inventory may introduce systematic biases, leading to overestimation or underestimation of posterior sectoral emissions in various regions. Specifically, in China, EDGAR and CAMS exhibit higher power generation sector emissions, whereas MEIC shows industrial sector emissions approximately 10% larger. This discrepancy may explain why emission changes from 2019 to 2020 estimated based on the MEIC inventory are dominated by industrial and transportation sectors, while this study identifies transportation, power generation, and industrial combustion as the principal drivers [20, 35, 51]. At the sub-regional scale, inventories show notable differences in power generation and transportation emissions in the North China Plain, and industrial, power generation, and transportation emissions in the northwestern region (Figure S23). In the contiguous U.S., CAMS power generation sector emissions exceed those in EDGAR by approximately 13%, and discrepancies in the East South Central region approach 20%, while EDGAR shows more prominent industrial emissions in the East South Central and West South Central regions, reflecting differences in how various inventories represent energy structure and industrial distribution (Figure S24). In Europe, CAMS and EDGAR exhibit pronounced divergences in the power generation and transportation sectors, with higher power generation emissions in CAMS and stronger transportation emissions in EDGAR (Figure S25). Although discrepancies exist among different emission inventories in the sectoral allocation fractions at the grid scale, the sectoral contributions exhibit a high degree of consistency across inventories at the large-scale level. Consequently, even if the sectoral fractions from other inventories are used to re-disaggregate the same total inversion-derived emissions, the qualitative identification of the dominant changing sectors (i.e., transport, industry, and power as the top three contributors) would remain unchanged, with differences manifesting in the quantitative percentages of absolute sectoral contributions. At the sub-regional scale, however, discrepancies in sectoral fractions among inventories can significantly alter local sectoral attribution. For example, over the North China Plain or the U.S. East South Central region, the estimated shares of the industrial versus power sectors may differ substantially among inventories, introducing uncertainty in identifying the dominant sector in such sub-regions. Nevertheless, these local-scale allocation uncertainties do not affect the core conclusions drawn from the large-scale overall pattern in this study. To mitigate such biases, future studies could adopt an ensemble-mean approach across multiple prior inventories to enhance the robustness of sector-scale attribution analyses.

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4. Conclusion

This study presents a top-down anthropogenic NO_x emission inversion framework based on the EnSRF data assimilation and GEOS-Chem, jointly assimilating TROPOMI satellite columns and surface NO₂ observations to estimate emissions over China, the contiguous U.S., and Europe during 2019–2020. The results indicate that anthropogenic NO_x emissions in 2020 decreased by approximately 5.38%, 8.94%, and 15.38% for China, the contiguous U.S., and Europe, respectively, relative to 2019. China's reductions were associated with the transportation, power, and industry sectors, whereas those in the contiguous U.S. and Europe were overwhelmingly dominated by the transportation sector, underscoring fundamental distinctions in industrial structure and energy consumption patterns between developed and developing economies. The inversion successfully captured the abrupt emission plunge during initial lockdowns, the subsequent recovery rebound, and the renewed decline during resurgent waves, temporal dynamics that prior bottom-up inventories based on fixed activity data severely underestimated. During the initial wave, emissions declined by 18.04%, 10.56%, and 21.74% in China, the contiguous U.S., and Europe, respectively. Emission changes exhibited pronounced spatial heterogeneity at the sub-regional scale, with the North China Plain and southeastern China, the Middle Atlantic and South Atlantic regions of the U.S., and Southern Europe experiencing the largest reductions. It should be noted that the inversion framework employed in this study directly constrains the spatiotemporal distribution and magnitude of anthropogenic NO_x emission changes from observed NO₂ concentrations. The sectoral contributions presented herein are derived by applying the posterior total-emission to the grid-scale sectoral fractions of the prior inventory. While the temporal evolution of these emissions overlaps with periods of COVID-19 related restrictions, the causal attribution of specific emission changes to particular lockdown measures, economic recovery patterns, or behavioral responses represents interpretive inference based on temporal consistency, rather than results directly constrained by the assimilation system. Future studies could incorporate independent data, such as traffic volume, electricity generation, and industrial output, to provide additional support for this inference.

In summary, this study provides quantitative evidence for the non-stationarity and regional specificity of NO_x emission dynamics under abrupt public health events, with important implications for validating bottom-up inventories and informing coordinated emission reduction policies.

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The TROPOMI NO₂ vertical column densities data are available at https://disc.gsfc.nasa.gov/datasets/S5P_L2_NO2_HiR_2/summary, the air quality data of China National Environmental Monitoring Centre (CNEMC) are available at <https://www.cnemc.cn/sss/>, the air quality data of the United States Environmental Protection Agency (EPA) are available at <https://www.epa.gov/outdoor-air-quality-data>, the air quality of the European Environment Agency (EEA) are available at <https://www.eea.europa.eu/en/datahub>.

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